The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Homomorphic Multi-Agent Lattices and Fitting Consensus

The architecture unifies the Epistemic Knowledge Lattice $\mathcal{L}_K$ and the Security Flow Lattice $\mathcal{L}_S$ into an identical branching partial-order structure governed by $a_k \in \mathcal{A}$. Parallel worker agents a_1 and a_2 occupy the same tier coordinate, formalizing a non-linear poset governed by the following strict axiomatic constraints:

$$a_3 \preceq a_2, \quad a_3 \preceq a_1, \quad a_2 \preceq a_0, \quad a_1 \preceq a_0 \tag{2}$$

4 Adversarial Logic Collisions and the Dialectical Argumentation Protocol

We formalize the mechanism for isolating and repairing conflicting agent ontologies during Model Context Protocol (MCP) theory movement passes.

4.1 Axiomatic Inconsistency Isolation

Let $\mathcal{K}_0$ represent the immutable master system invariants stored within the apex node a_0 (`agent_0_master`). A local theory extension $\Delta\mathcal{K}_1$ passed by a source agent a_1 is classified as a consistent local logic

extension if and only if the joint set is completely regular with respect to the core distributed logic channel:

$$\mathcal{K}_0 \cup \Delta\mathcal{K}_1 \nvdash \bot \tag{3}$$

If a hostile mutation occurs such that the source agent asserts a rule contradicting a core invariant (e.g., $\Delta\mathcal{K}_1 \vdash$ allow_data_leaks = True while $\mathcal{K}_0 \vdash$ allow_data_leaks = False), a hard contradiction boundary is breached ($\mathcal{K}_0 \cup \Delta\mathcal{K}_1 \vdash \bot$). The target agent immediately triggers Rasiowa's Clause (e) Modal Negation, dropping the packet and issuing a `REJECT_CONFLICTING_THEORY` status token.

4.2 The Argumentation Protocol as a Self-Repair Loop

To resolve the contradiction without catastrophic system halt, an out-of-band Argumentation Protocol is initiated over the MCP tool layer. The target agent transmits a formal symbolic counter-proof $\mathcal{P}_{\text{audit}}$ derived from the apex invariant base back down the poset link to the corrupted source node:

$$a_2 \xrightarrow{\mathcal{P}_{\text{audit}}} a_1 \quad \text{where} \quad \mathcal{P}_{\text{audit}} \implies (\text{allow_data_leaks} \equiv \text{False}) \tag{4}$$

The protocol forces the source agent a_1 to execute a local corrective transaction pass:

1. **Logical Overwrite:** The source node rewrites its local `kb.pl` file to match the audited counter-proof.
2. **Forest Back-Propagation Repair:** The source node executes a recursive tree-pruning pass across its local decision forest $\mathcal{F}_1$, clearing any fraudulent reward leaf nodes where the exploit was assigned a positive weight.

Once the local correction pass achieves algebraic closure, the theory migration retries, culminating in a successful, zero-uncertainty Graph-of-Trees merge.